Lecture 12: Outline

- A few element of stochastics:
  - random variables,
  - expected value, variance.
  - random vectors and matrices
  - two important examples.

Lecture 12: Motivation

- Main question: How this comes into the picture?
- Example: Deep learning-based weather prediction.
- Main task: learn the function

  F : recent data $\to$ future data.

  - recent data: a history of measured data
- Learning data: past observation pairs.
- Observations are polluted with random errors.
  - With formulas:

    $F : X_{\text{recent}} \to X_{\text{future}}$ – learn this "correct prediction"
  - $(X_{\text{recent, obs}} + \text{Err}_1, X_{\text{future, obs}} + \text{Err}_2)$ – learning data
    - We do not have "exact" learning data.

Lecture 12: Programming with stochastics

- Efficient subroutines in Python (we use this).
- Statistics toolbox in Matlab.
- R language: similar, specialized to this topic.
- Visualization tools in each language.

Lecture 12: Random variables

- Informally: a function with random (function)values.
- A bit more precise for real-valued random variables:
  - A random variable is a function $X : \Omega \to \mathbb{R}$, where
  - Ω is the set of all possible outcomes in some experiment.
- Example 1; Lottery:
  - $\Omega = \{\omega : \omega = \text{choice of a ball}\}$
  - $X(\omega)$: number on the ball
- Example 2; Measuring height:
  - $\Omega = \{\omega : \omega = \text{choice of a person}\}$
  - $X(\omega)$: height of this person.

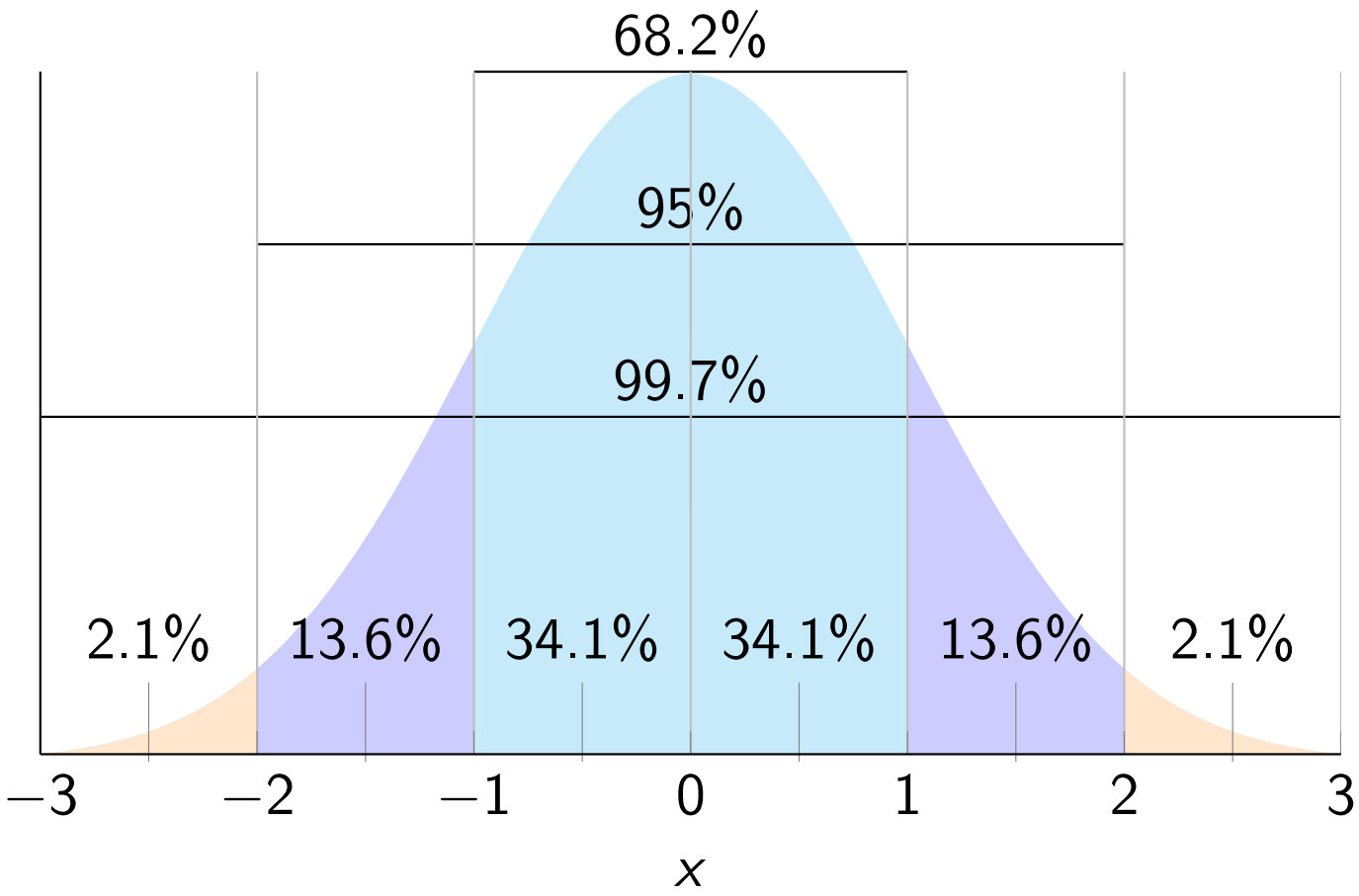
Lecture 12: Characterizing random variables I.

- Probability: P - a function,
  - defined on certain subsets of Ω,
  - Additive: $A_1 \cap A_2 = \emptyset \Rightarrow P(A_1 \cup A_2) = P(A_1) + P(A_2)$.
    - Even for countably infinity number of sets.
  - mapping them to $[0, 1]$; $P(\Omega) = 1$.
- Example 1; Lottery:
  - $P(\{\omega : X(\omega) = 28\}) = \frac{1}{90}$, $P(\{\omega : X(\omega) \leq 45\}) = \frac{1}{2}$.
  - For finite number of outcomes, this is enough to compute everything.
- Example 2; Measuring height:
  - $P(\{\omega : X(\omega) = 183.4\,\mathrm{cm}\}) = P(X = 183.4) = 0$,
    $P(\{\omega : X(\omega) \geq 190\,\mathrm{cm}\}) = P(X \geq 190) = \frac{1}{25}$.
  - For infinite number of outcomes, some other quantity is necessary for computations.

Lecture 12: Characterizing random variables II.

- Density function of X; notation: f_X
  - $f_X(x) = \lim_{\delta \to 0} \dfrac{P(x - \delta < X \leq x + \delta)}{2\delta}$, if this makes sense.
  - In this case: X is called continuously distributed.
- Intuitive explanation:
  - This is "large" around x : values around x can be observed with "large" probability.
  - In general: $P(a < X \leq b) = \int_a^b f_X(x)\,\mathrm{d}x$.
  - Of course: $1 = P(-\infty < X \leq \infty) = \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x$.

Lecture 12: A density function visually

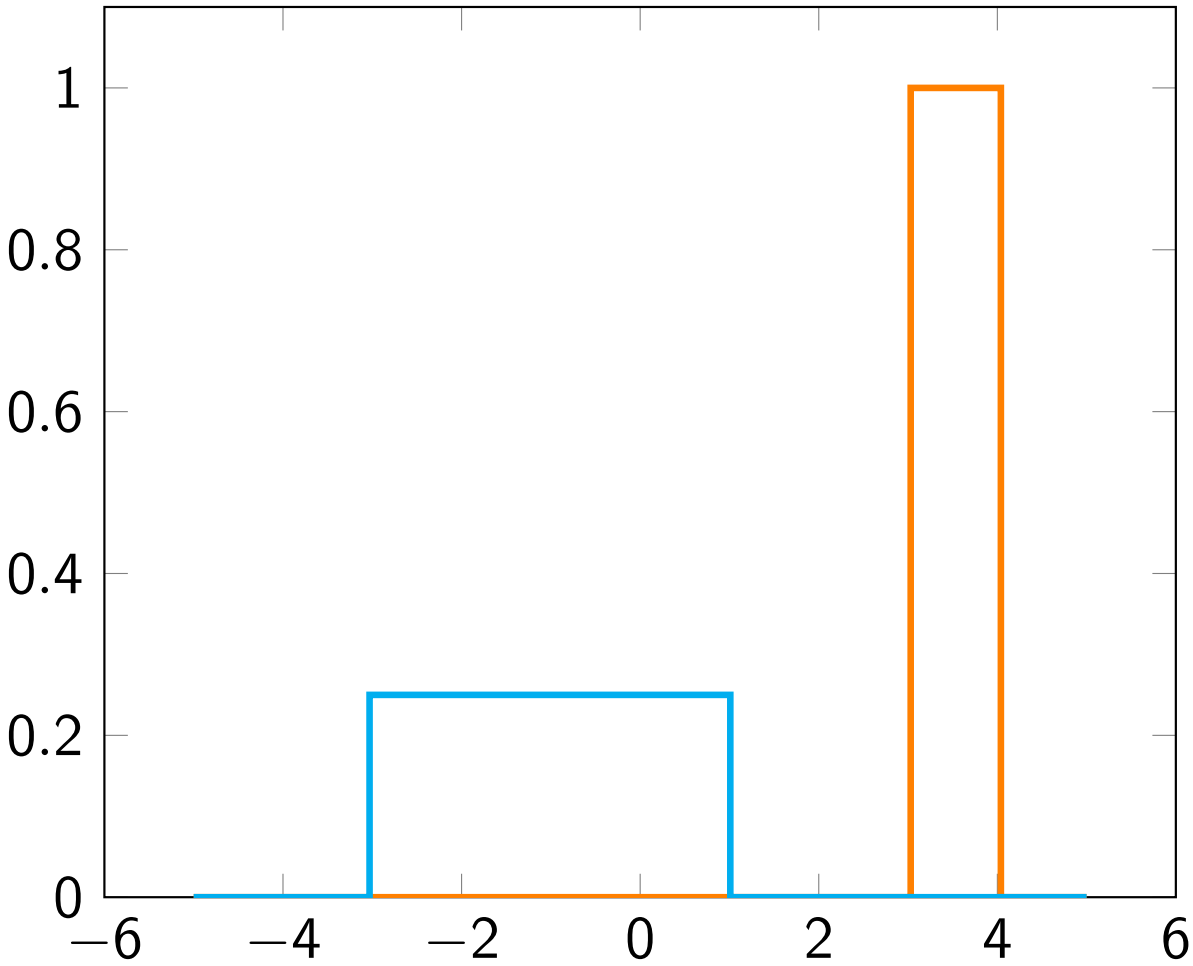


Lecture 12: Random variables - an important example

- Example 1: X is uniformly distributed on $[a, b]$.
  - $f_X(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b], \\ 0 & x \notin [a, b]. \end{cases}$
  - Now values can occur only in $[a, b]$.
  - Equal chance for possible numbers: constant f in $[a, b]$.
  - $P(x \in I)$ is proportional with the length of I.
  - Expected value of X: $\frac{a+b}{2}$.
  - Variance of X: $\frac{(b-a)^2}{12}$.
- "Random number" in Python: uniformly distributed in $[0, 1]$.
- "Random vector, matrix" in computers: vectors, matrices with the above independent components.

Lecture 12: Two uniform density functions

- For $[a, b] = [-3, 1]$ (blue) and $[a, b] = [3, 4]$ (orange)



Lecture 12: Random variables - another important example

- Example 2: X has normal distribution; $X \sim N(m, \sigma)$
  - Then $f_{m,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$ with $m \in \mathbb{R}, \sigma \in \mathbb{R}^+$. Very informally:
    - Now values can occur anywhere on $\mathbb{R}$.
    - Values far away from m can be observed with very small probability.
  - Further properties:
    - Expected value of X is m,
    - Variance of X is σ^2.
  - X has *standard normal* distribution:

    if $m = 0, \sigma = 1$, i.e., $f_{0,1}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

Lecture 12: Normally distributed random variables: importance

- Occurrence 1: in a majority of noises/ in measurements, which are added to the “real”, deterministic data.
  - Measurement of position (GPS data).
  - Recording of speech (for recognition).
- Occurrence 2: in some biological data
  - Height in a fixed (adult, men) population.
  - Birth weight in a fixed population.
- Non-occurrence: We want simply to “randomize" a data set by adding noise.
  - Test the robustness of some computational method.

Lecture 12: Transformation of random variables I.

- If X is uniformly distributed on $[a, b] \subset \mathbb{R}^+$, then for any $K > 0$:
  - $K + X$ is uniformly distributed on $[a + K, b + K]$
  - $K \cdot X$ is uniformly distributed on $[Ka, Kb]$
- Be careful: if X_1 and X_2 are independent, uniformly distributed on $[a, b] \subset \mathbb{R}^+$,
  - $X_1 + X_2$ is *not* uniformly distributed on $[2a, 2b]$.

Lecture 12: Transformation of random variables II.

- If X has a standard normal distribution, then for any $K > 0$:
  - $K + X$ is also normally distributed with $f_{K+X}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-K)^2}{2}}$
  - $m + \sigma X$ is also normally distributed with $f_{m,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$
- We can transform it back: standardize the random variable X:

  $X \sim N(m, \sigma) \Rightarrow \frac{X-m}{\sigma} \sim N(0, 1)$
- Now if X_1 and X_2 are independent, normally distributed, their sum $X_1 + X_2$ is also normally distributed.

Lecture 12: Transformation of random variables III.

- In principle, it would be enough to generate only one random (e.g., uniformly distributed) data $y_1, y_2, \ldots, y_k$.
  - For a density function f_X take the cumulative distribution function: $F_X(x) = \int_{-\infty}^{x} f_X(s)\,\mathrm{d}s$,
  - Transform the above data by

$$F_X^{-1}(y_1), F_X^{-1}(y_2), \ldots, F_X^{-1}(y_k).$$

  - This is distributed according to f_X.
- There are specialized (more efficient) algorithms for certain probability distributions (e.g., for the normal), which were implemented in Python.

Lecture 12: Standardization of random variables I.

- The transformation normal $\rightarrow$ standard normal in general framework.
- Motivation: in many cases, observed data can be considered as a random variable (data = "real value" + noise)
  - Would be fine but we do not know expected value and variance a priori.
  - OK, instead, we use their estimates.
- `StandardScaler` in Python:
  - Instead of $x_1, x_2, \ldots, x_n$, we use

$$\frac{x_1 - \overline{x}}{\sigma_x}, \frac{x_2 - \overline{x}}{\sigma_x}, \ldots, \frac{x_n - \overline{x}}{\sigma_x},$$

    where $\overline{x}$ – average, σ_x – standard deviation.

Lecture 12: Standardization of random variables II.

- Another option: `MinMaxScaler` in Python
  - We transform $x_1, x_2, \ldots, x_n$ into

$$\frac{x_1 - \overline{x}}{\max_j x_j - \min_j x_j}, \frac{x_2 - \overline{x}}{\max_j x_j - \min_j x_j}, \ldots, \frac{x_n - \overline{x}}{\max_j x_j - \min_j x_j},$$

- In real-life situations, these are still not enough: a pre-filtering of data is necessary. Especially min and max are rather unreliable.

Lecture 12: Vector-valued random variables

- In practice: we measure more quantities simultaneously and try to explore their relation. Mathematically; we measure a vector variable.
  - An important case:

    Two random variables X and Y are said to be *bivariate normal*, or jointly normal, if $aX + bY$ has a normal distribution for all $a, b \in \mathbb{R}$.
  - Example: In a fixed population, we measure the daily energy intake of people and the weight of the same people.

    Each observation corresponds to an individual and has two components.
- How to describe such a distribution?
- In any case, we should yield the relation of the components in the vector.

Lecture 12: Bivariate normal distribution $\mathbf{X}$

- We again use a density function: $f : \mathbb{R}^2 \to \mathbb{R}$ such that

  $$P(\mathbf{X} \in U) = \int_U f(\boldsymbol{s})\,\mathrm{d}\boldsymbol{s}.$$

- This is defined with

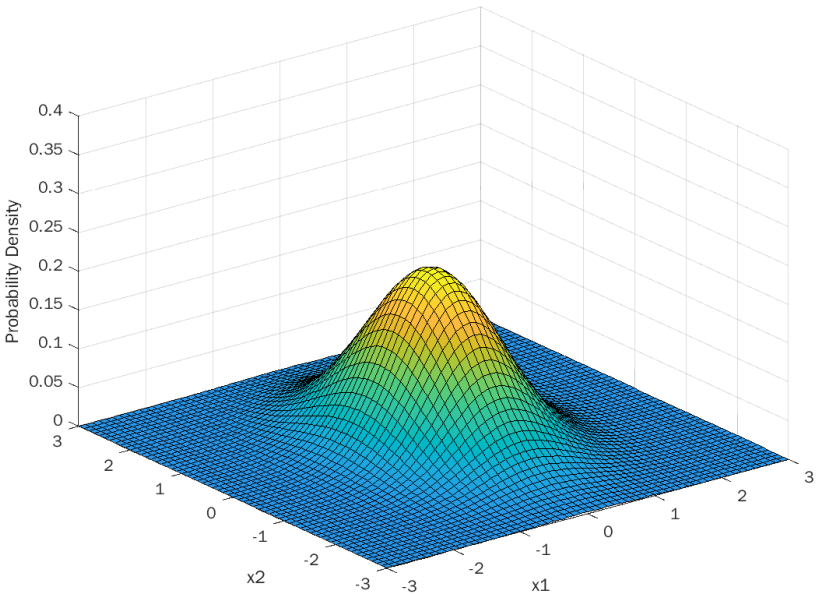
  $$f(\boldsymbol{x}) = \frac{1}{2\pi\sqrt{\det \Sigma}} \cdot \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \cdot \Sigma^{-1} \cdot (\boldsymbol{x}-\boldsymbol{\mu}),\right\}$$

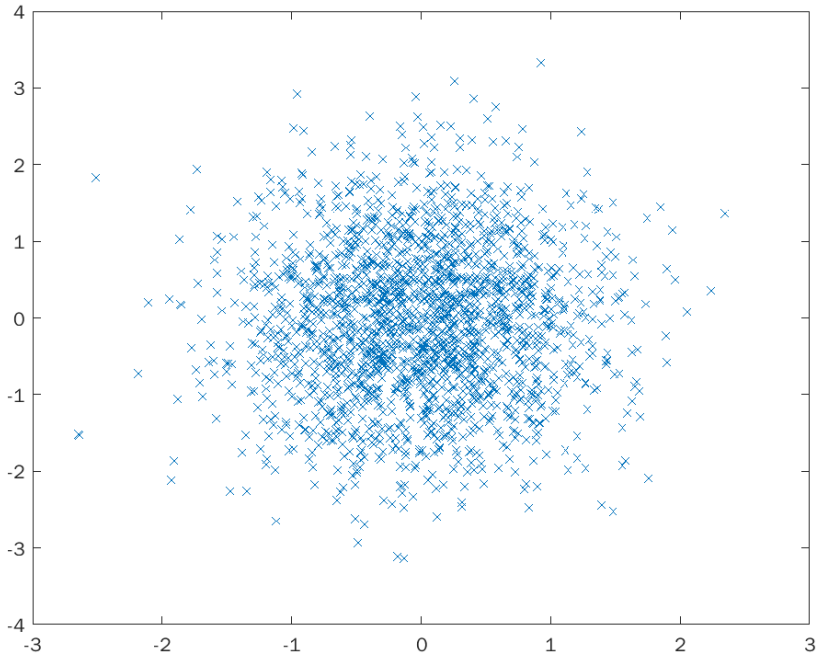
  where

  - $\boldsymbol{\mu}$ is the expected (vector) value,
  - Σ is the covariance matrix (measured between the components).

Lecture 12: Bivariate normal distribution $\mathbf{X}$, example 1

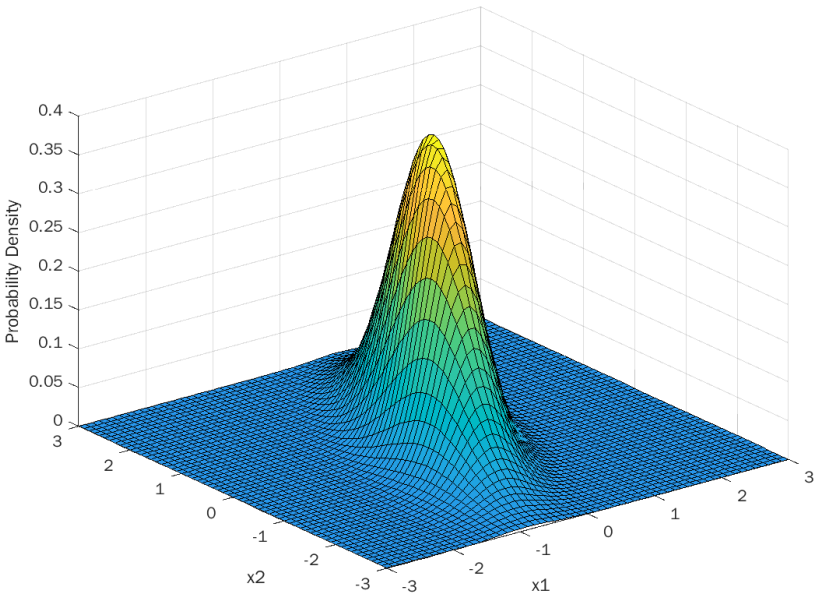
- Example 1: $\boldsymbol{\mu} = (0, 0)$, $\Sigma = \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix}$ - uncorrelated (independent) data.
- Density function (left), a random sample of size 2000 (right).

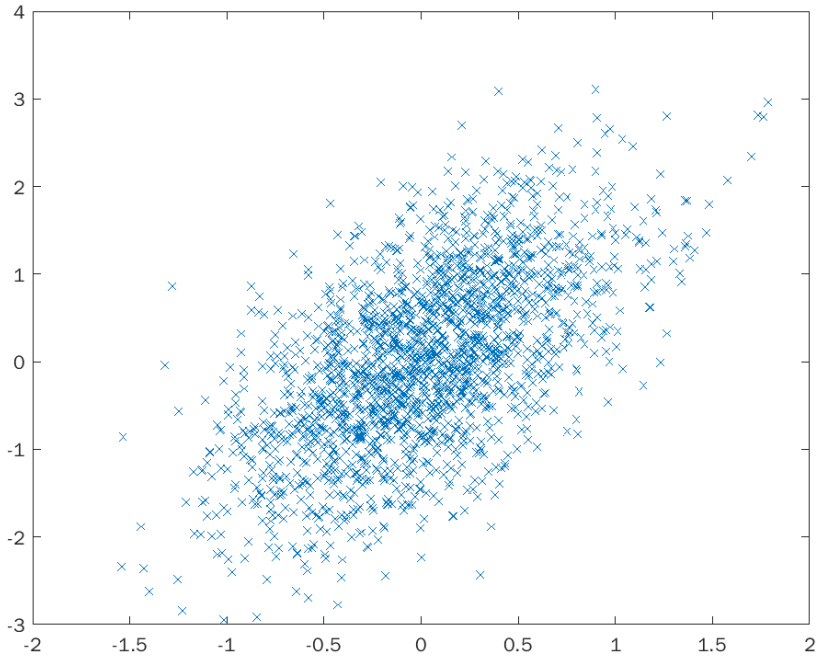




Lecture 12: Bivariate normal distribution $\mathbf{X}$, example 2

- Example 1: $\boldsymbol{\mu} = (0, 0)$, $\Sigma = \begin{pmatrix} 0.25 & 0.3 \\ 0.3 & 1 \end{pmatrix}$ - correlated (dependent) data.
- Density function (left), a random sample of size 2000 (right).





Lecture 12: Bivariate normal distribution: explanation of parameters

- $\Sigma[1,1]$: variance of the first component
- $\Sigma[2,2]$: variance of the second component
- $\Sigma[1,2] = \Sigma[2,1] =:$ covariance of the first and second components
- Contour lines of the density function: ellipses.
- Eigenvectors of Σ: direction of the principal axes of the above ellipses.
  - OK, they are real,
  - they are orthogonal to each other.